Proteomics Analysis Using the JMod GUI

While JMod can be run from the command line, many users may find it easier to use a graphical user interface. Here, we walk through the important aspects of the user interface and how it can be used to analyze data with JMod.

Installation:

JMod can be downloaded as an executable for hyperlink[windows] or hyperlink[mac], or found on GitHub.

If using windows, it is recommended to enable long paths to ensure all outputs are properly saved. To do this:

1) Press Win+R
2) Type ‘regedit’ and hit enter
3) Navigate to HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\FileSystem
4) Set LongPathsEnabled to 1
5) Restart computer

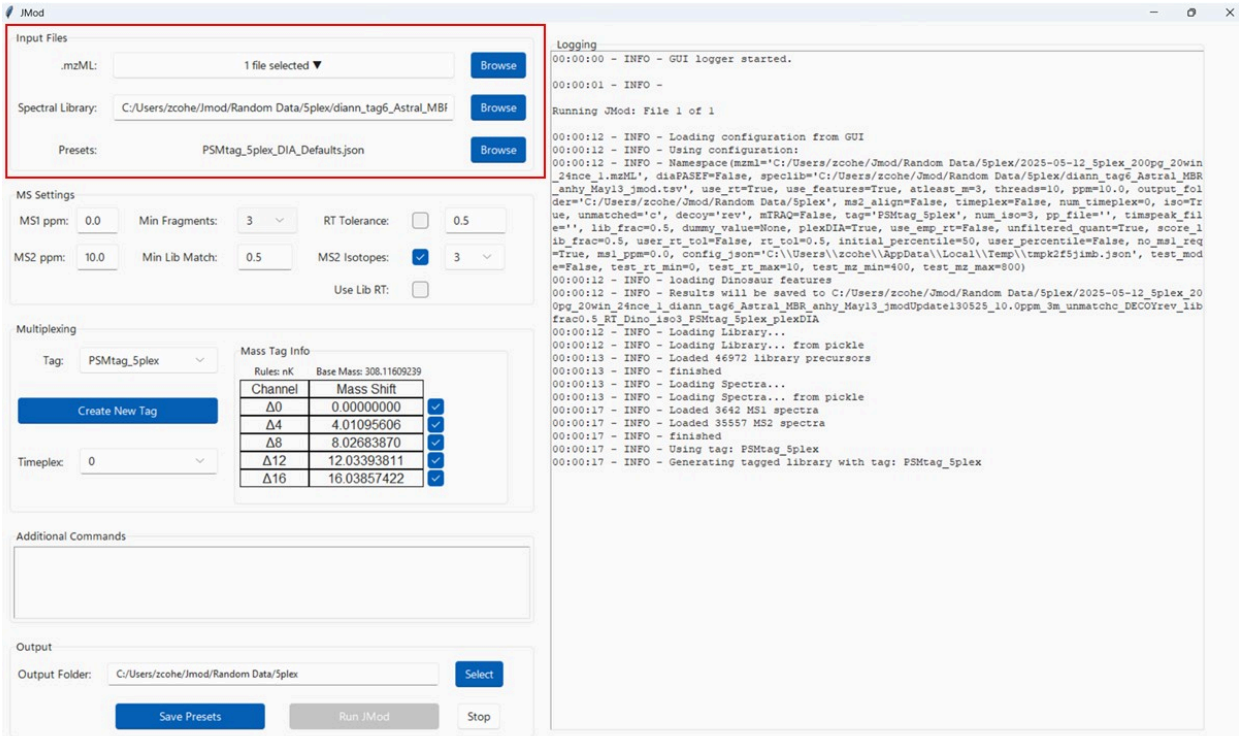
Processing the data:

JMod requires mzML files as inputs. For conversion from Thermo .raw files, we recommend using msConvert with parameters:

- Peak Picking = 1-
- The rest are default

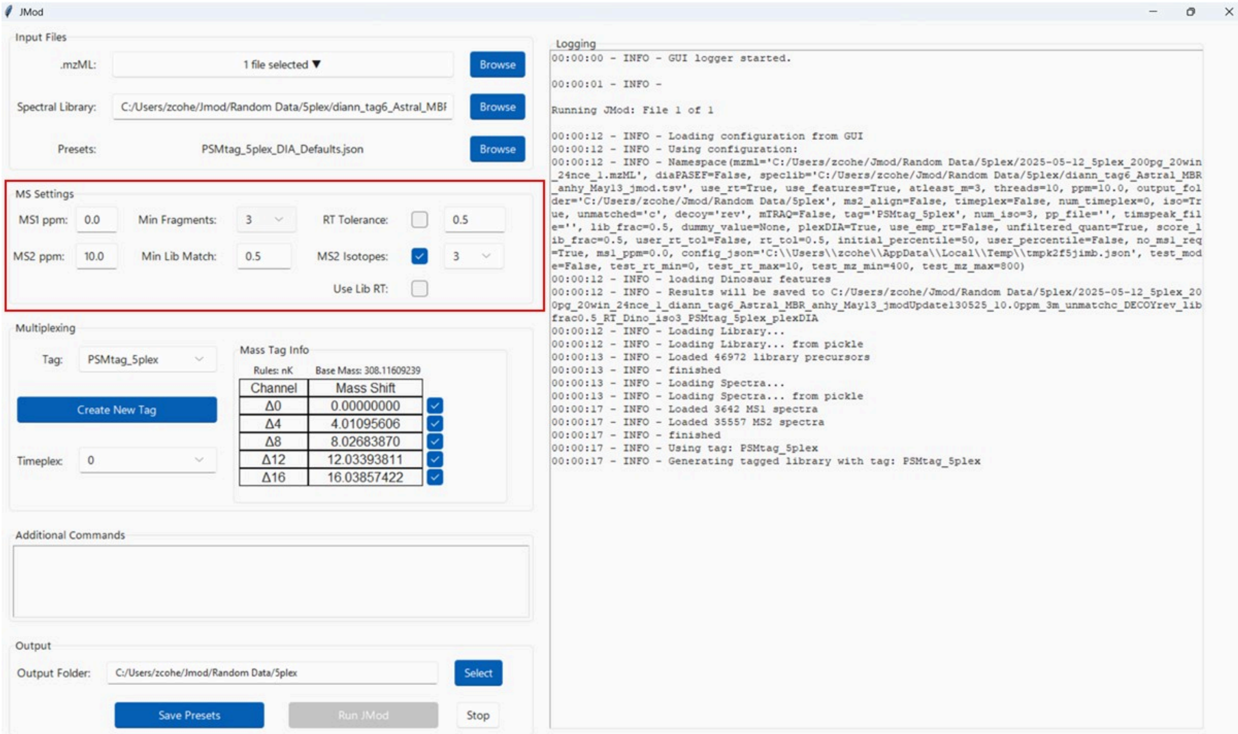
Working with the GUI:

Step 1: Input mzML file(s) and spectral library and select presets



JMod takes one or multiple mzML files as input. JMod requires a spectral library for use. For multiplexed data with nonisobaric mass tags (such as PSMtag), the spectral library should contain peptides labelled with the lowest mass isotopologue (or “d0”). We recommend using empirical spectral libraries for optimal performance. For more on generating PSMtag libraries, check out [this blogpost](). Optionally, a configuration file can be chosen from among a set of presets, or from past JMod runs. Configuration files can be found in the output folder of all JMod searches.

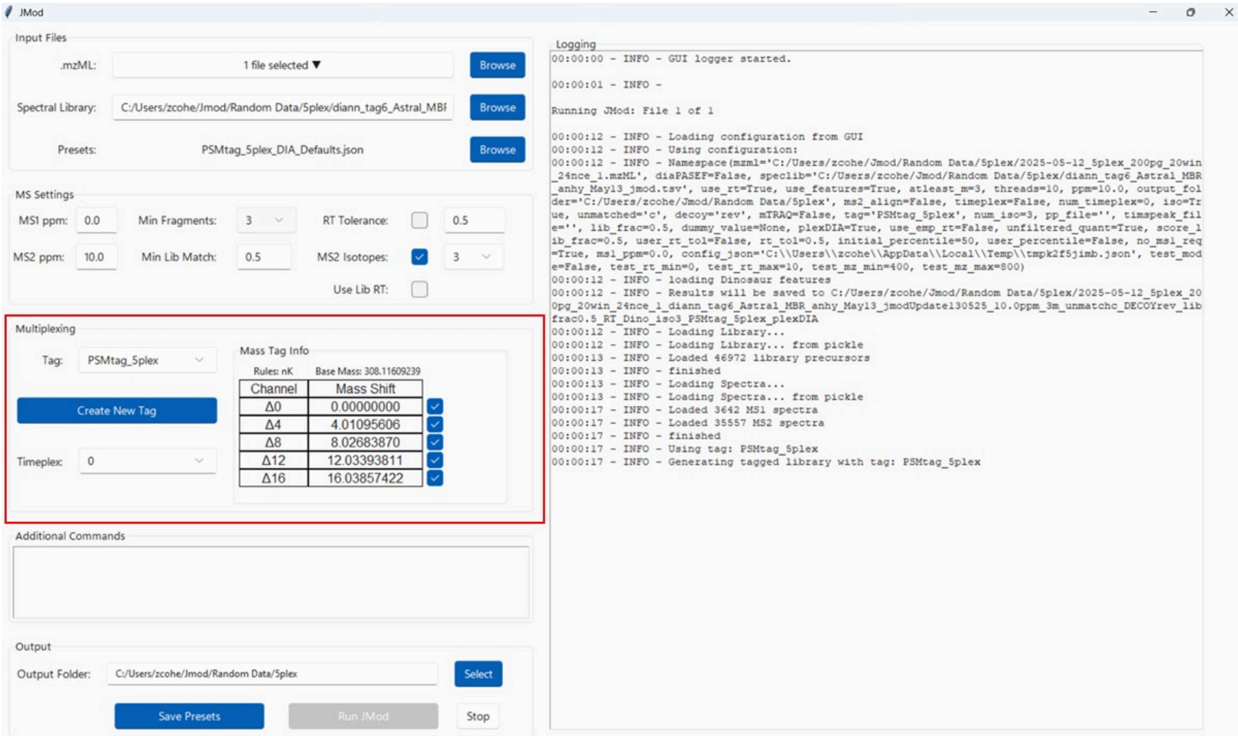
Step 2: Selecting MS related parameters



Various parameters of the JMod search can be modulated from the GUI. These include:

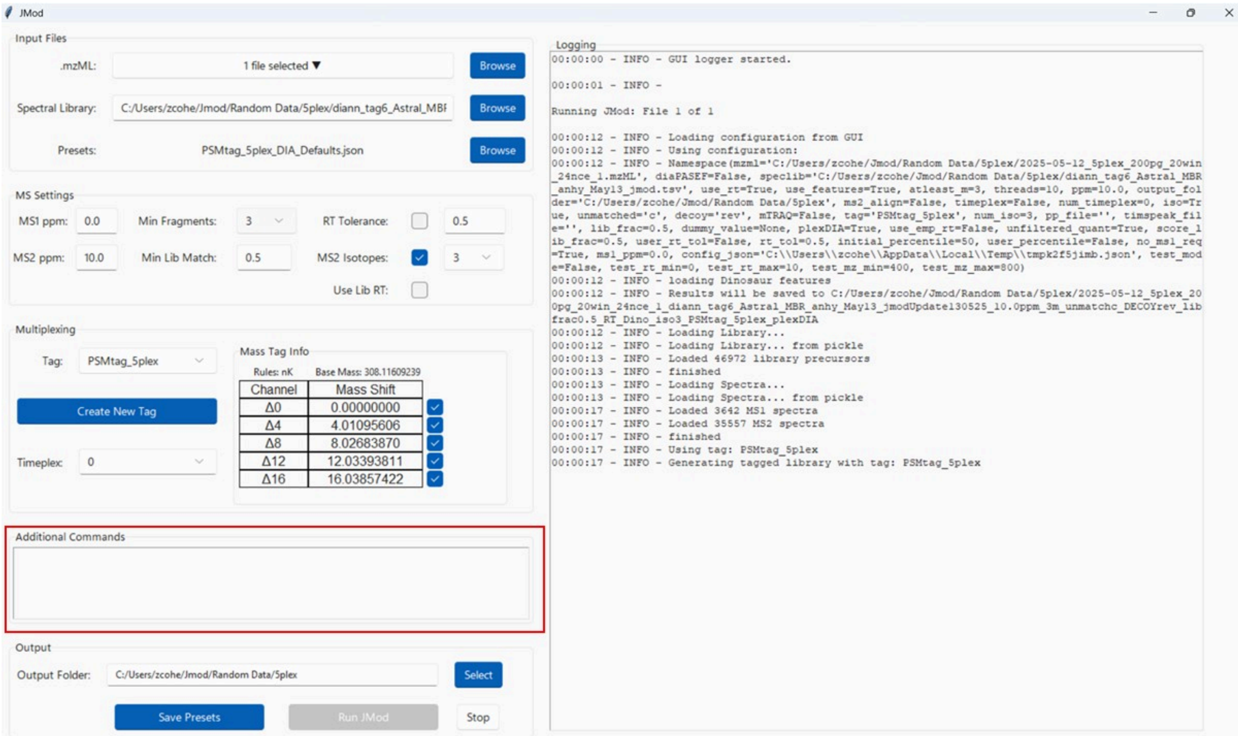
```
MS1 ppm          :  MS1 matching tolerance in parts-per-million.
                     If 0, optimized value will be calculated
MS2 ppm          :  MS2 matching tolerance in parts-per-million
Min Fragments    :  Minimum number of fragments matched for
                    precursor identification.
Min Lib Match    :  Minimum fraction of library intensity
                     matched for precursor to be considered
RT Tolerance     :  If turned on, set an explicit retention time
                     tolerance to be used.
MS2 Isotopes     :  If turned on, the number of isotopes used
                     to search MS2 spectra
Use lib RT       :  Use library retention time instead of
                     fine-tuning retention time
```

Step 3: Selecting Parameters for Multiplexing



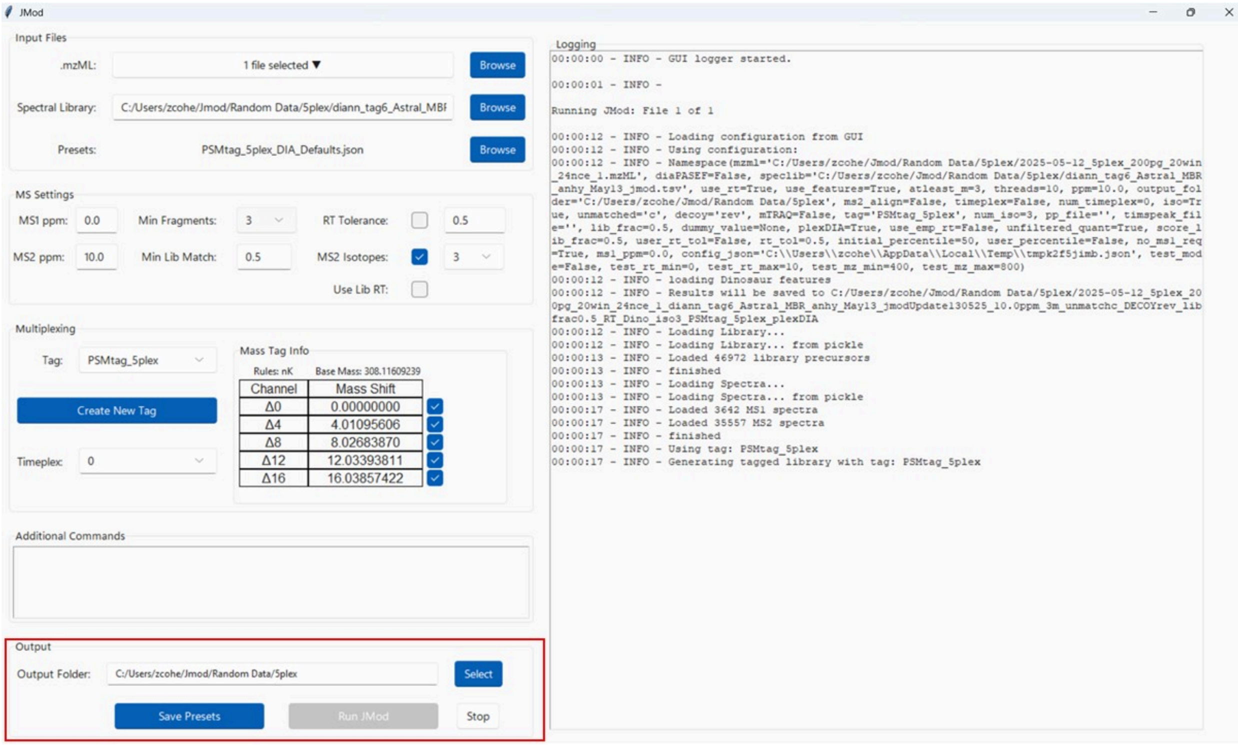
Next, if multiplexing, either with nonisobaric mass tags or with timeplex, parameters for these can be selected. From a dropdown menu, select from premade nonisobaric tags, including diethyl, mTRAQ, and PSMtag. For label free data, set the tag to “None”. The channels and mass shifts for the tag will be displayed to the right. By selecting or unselecting the boxes next to each channel, data can be searched with this subset of channels only. Additionally, other tags can be created for the search with the “Create New Tag” button. Once created, these will populate in the tag dropdown menu for this run, as well as subsequent runs. If using timeplex, the number of time channels in use can be selected from the timeplex dropdown menu.

Step 4: Additional Commands



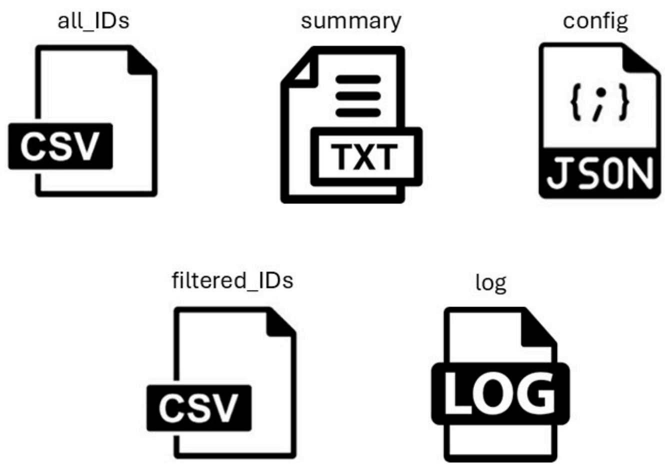
While the GUI does not have a widget for every possible command line argument, these arguments can be written exactly as they are in the command line in the entry here and will be parsed the same way.

Step 5: Selecting Output and Running



Select a folder for outputs to be placed in. Each .mzML file will have a unique folder within this containing its specific outputs. A configuration file for the current parameters can be saved beforehand using “Save Presets”. However, this configuration file will also be created automatically in the output folder. Selecting run will perform sequential JMod searches on each selected .mzML. Progress can be monitored via the log output on the right-hand side of the GUI, and outputs from each file are available as soon as it is finished running.

Step 6: Looking at Jmod Outputs



JMod outputs 5 important files (along with many others that can be used for quality control). Tabular results are contained in **all_IDs.csv** and **filtered_IDs.csv** (with filtered_IDs showing only peptides with Q < 0.01). Columns for these are described hyperlink[here]. **Summary.txt** shows the number of proteins and precursors identified at 1% FDR, and for plexDIA additionally shows the number of precursors and proteins where the best channel is identified at 1% FDR. Additionally **Log.log** contains logged statements from the run, and **config.json** contains the parameters of the run, which can be used as a “preset” in the GUI, or to run a search from the command line.